FRAMEWORK FOR DATA AND VISUAL ANALYTICS
MINI PROJECT
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Abstract:

The Sales Performance Analysis and Dashboard project aims to analyze sales data and present meaningful business insights using data visualization tools such as Microsoft Power BI and Tableau. The project uses a dataset containing 10,000 sales records with details such as order ID, order date, region, city, product category, quantity, sales amount, profit, order status, sales channel, and payment method.

The dataset is imported into Power BI and Tableau to create interactive and informative visualizations. In Power BI, key performance indicators (KPIs) such as Total Sales, Total Profit, Total Orders, and Total Quantity are calculated and displayed using card visuals. Charts are created to analyze sales performance by region, monthly sales trends, product categories, and order status. Similarly, Tableau is used to develop a dashboard containing four visualizations: Sales by Region, Monthly Sales Trend, Sales by Category, and Order Status.

The dashboards help users understand sales patterns, compare regional performance, identify profitable categories, and examine the distribution of order statuses. By converting raw sales data into graphical representations, the project makes information easier to understand and supports data-driven decision-making.

This project demonstrates the practical application of business intelligence and data visualization techniques in sales analysis. The developed dashboards can help business users monitor performance, identify trends, and make informed decisions to improve sales and profitability.

Objectives:

1. To analyze sales data containing 10,000 records using Power BI and Tableau.
2. To calculate important performance indicators such as total sales, total profit, total orders, and total quantity.
3. To visualize regional sales performance using bar charts.
4. To analyze monthly sales trends using line charts.
5. To compare sales across different product categories using pie or donut charts.
6. To represent the distribution of order statuses through graphical visualizations.
7. To develop clear and user-friendly dashboards for effective data interpretation.
8. To support business decision-making by identifying sales patterns and performance trends.

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Implementation:

Dataset Collection:

A sales dataset containing 10,000 records is collected in CSV format. It includes information about orders, dates, regions, products, quantities, sales, profits, and order statuses.

Data Import:

The CSV dataset is imported into Microsoft Power BI and Tableau Desktop. The data fields are examined to ensure that they can be used for analysis and visualization.

Data Preparation:

The dataset is checked for appropriate column types and usability. Date, numerical, and categorical fields are identified for creating the required visualizations.

Dashboard Development in Power BI

The following measures are created using DAX:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity

Card visuals are used to display the KPI values. A bar chart is created to compare sales by region, a line chart to show monthly sales trends, and donut charts to represent sales by category and order status.

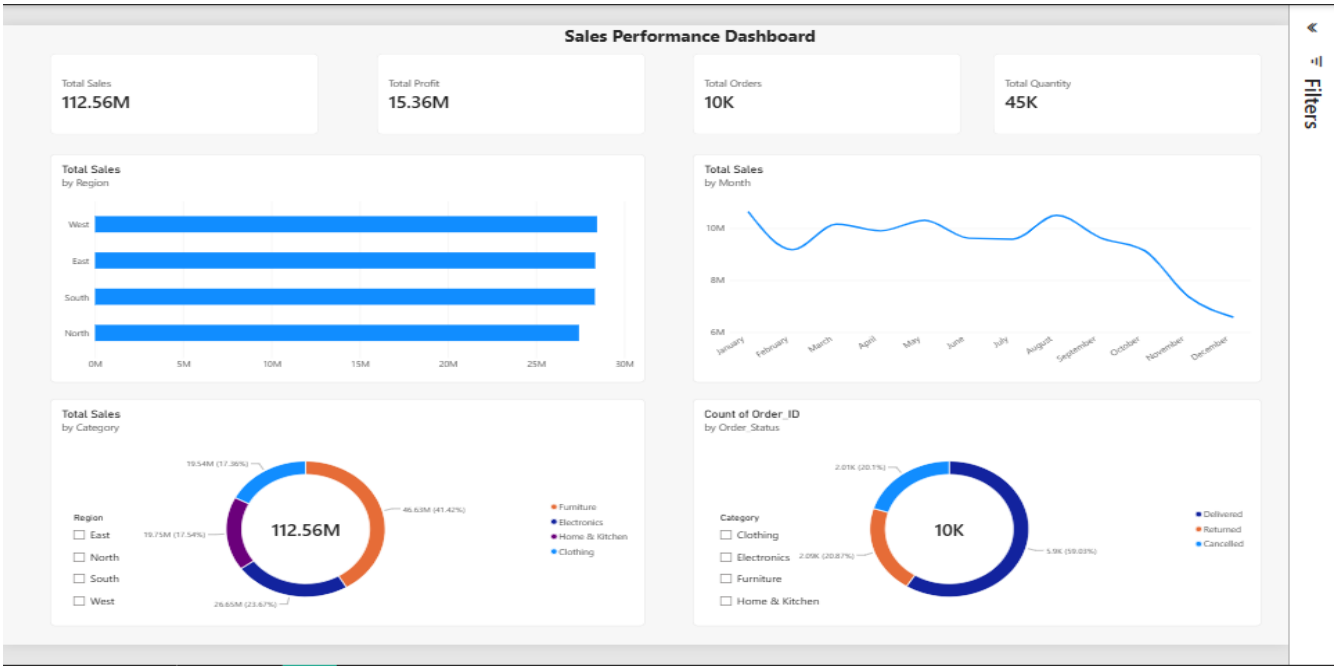
Dashboard Development in Tableau:

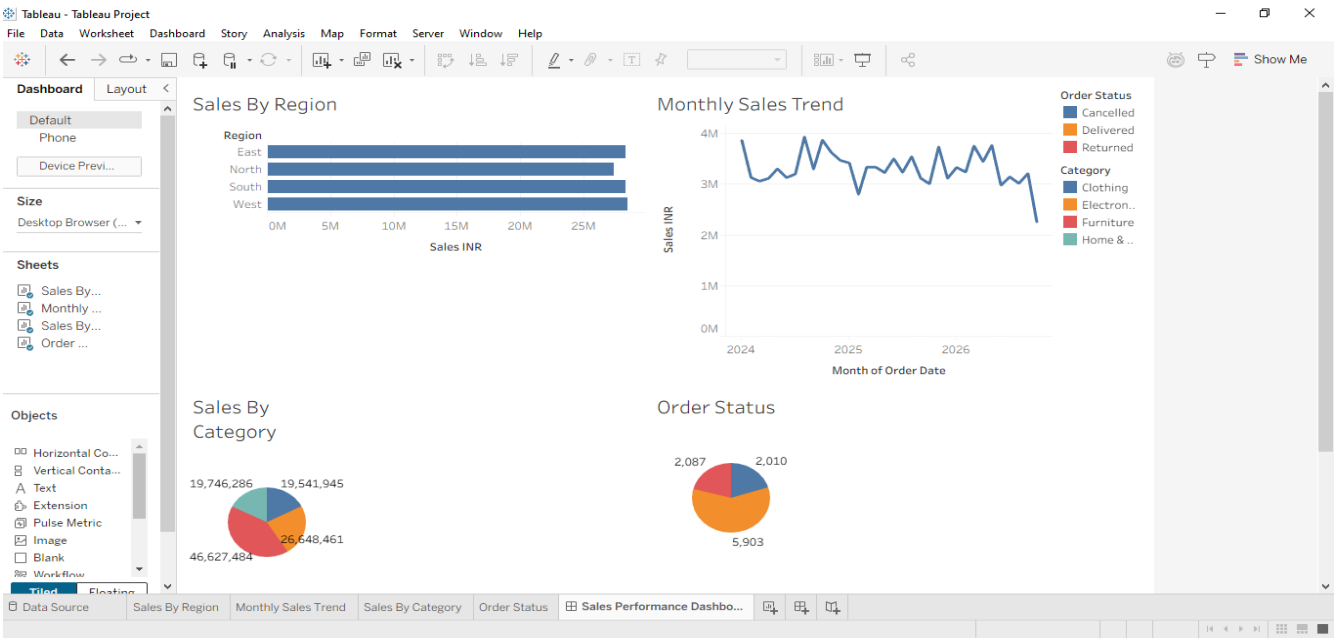
The same sales dataset is connected to Tableau Desktop. Four worksheets are created:

- Sales by Region — a bar chart comparing sales across regions.
- Monthly Sales Trend — a line chart showing changes in sales over time.
- Sales by Category — a pie chart representing sales distribution across product categories.
- Order Status — a pie chart displaying the distribution of orders by status.

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Output:





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Conclusion:

The Sales Performance Analysis and Dashboard project successfully demonstrates how Microsoft Power BI and Tableau can be used to transform raw sales data into meaningful visual information. Using a dataset of 10,000 records, the project presents important sales indicators and visualizes regional performance, monthly sales trends, product category distribution, and order status.

The Power BI dashboard provides a summary of key performance indicators along with multiple charts, while the Tableau dashboard combines four worksheets into a single visual reporting interface. These dashboards make sales information easier to understand and help users identify patterns and compare business performance.

The project provides practical experience in data importing, data preparation, calculation of measures, chart creation, and dashboard development. In the future, the system can be enhanced by adding interactive filters, profit margin analysis, year-over-year comparisons, and sales forecasting techniques. Overall, the project demonstrates the importance of data visualization in business intelligence and data-driven decision-making.

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